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ERTMS/ETCS RAMS Requirements Specification

Chapter 0 - Glossary Acronyms References

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- Chapter 0 - Glossary Acronyms References -
- Chapter 1 - General Aspects 01s1266-
- Chapter 2 - RAM Requirements Specification 02s1266-
- Chapter 3 - Safety Requirements Specification 03s1266-

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0.1. NORMATIVE REFERENCES

EN 29000-3	"Quality Management and Quality Assurance Standards - Guidelines for the Application of ISO9001 to the development, supply and maintenance of software", CEN, First Version, June 1993
EN ISO 9000-1	"Quality Management and Quality Assurance Standards - Guidelines for Selection and Use", CEN, supersedes EN 29000, July 1994
EN 50126	"Railway Applications - The Specification and Demonstration of Dependability, Reliability, Availability, Maintainability and Safety (RAMS)", CENELEC
EN 50128	"Railway Applications - Software for Railway Control and Protection Systems", CENELEC
ENV 50129	"Railway Applications - Safety Related Electronic Systems", CENELEC
EN 50159-1	" Railway Applications - Requirements for Safety-Related Communication in Closed Transmission Systems", CENELEC
EN 50159-2	"Railway Applications - Requirements for Safety-Related Communication in Open Transmission Systems", CENELEC

0.2. MANDATORY REFERENCES

EEIG ERTMS USERS GROUP	"Engineering Documents and Eurosig Documents", See the list in Contents.xls CDROM 31/07/98
EEIG ERTMS USERS GROUP	"ERTMS - Quality Requirements for Suppliers", Version 1-, 20/09/96
EEIG ERTMS USERS GROUP	"ERTMS - Validation Procedures", Version 3-
EEIG ERTMS USERS GROUP	"ERTMS/ETCS - Environmental conditions", Version 5-
EEIG ERTMS USERS GROUP	"ERTMS Control/Command Test tool characteristics requirements for Software Safety Test", Version 2-
EEIG ERTMS USERS GROUP	" The Attribution of the ERTMS SRS Functions to Constituents", Version 1-
UIC/ERRI	"ERTMS - Requirement Specifications: Functional Requirements Specification FRS, Synopsis", Final Version, January 1996
UIC/ERRI	"ETCS - FRS Functional Requirements Specification", Version 4.01

0.3. INFORMATIVE REFERENCES

EEIG ERTMS USERS GROUP	RAMS Requirements - Informative Part 98s7111-
EEIG ERTMS USERS GROUP	"ERTMS/ETCS - Specification of Service Requirements", included in 98s7111- above
UIC/ERRI	"ETCS RAM Strategy", Final Draft, 28 July 1995
UIC/ERRI	"ETCS Safety Strategy" Final Draft, 31 December 1995
IEC 1508	"Functional Safety: Safety-Related Systems", IEC SC65A, Draft, June 1995
ISO/IEC DIS 9126	"Information Technology - Software Product and Evaluation - Quality Characteristics and Guidelines for their Use", ISO/IEC JTC-1, Draft, 1990
MIL-HDBK-338-1A	"Electronic Reliability Engineering Handbook", Vol. 1, USA DoD, 12 October 1988
MIL-HDBK-472	"Maintainability Prediction - Handbook", USA DoD, 24 May 1966
MIL-STD-1388-1A	"Logistic Support Analysis", USA DoD, 11 April 1983
MIL-STD-1629A	"Procedures for Performing a Failure Mode, Effects and Criticality Analysis", USA DoD, 24 November 1980
MIL-STD-470B	"Maintainability Program for Systems and Equipment", USA DoD, 30 May 1989
MIL-STD-471A	"Maintainability Verification/Demonstration/Evaluation", USA DoD, 27 March 1973
MIL-STD-721C	"Definitions of Terms for Reliability and Maintainability", USA DoD, 12 June 1981
MIL-STD-756B	"Reliability Modelling and Prediction", USA DoD, 18 November 1991
MIL-STD-781D	"Reliability Testing for Engineering Development, Qualification and Production", USA DoD, 17 October 1986
MIL-STD-785B	"Reliability Program for System and Equipment Development and Production", USA DoD, 15 September 1980

CENELEC SC9XA Ad Hoc	HAZARDOUS FAILURE RATES AND SAFETY LEVELS „Definition, methodology and figures to achieve cross-acceptance in Europe“, meeting in Marseille in June 1996
JAR 25	Joint Airworthiness Requirements, JAR 25, Large Aeroplanes, Section 1309, equipment, systems and installation.
Book	Michael R. Lyu, "Handbook of Software Reliability Engineering", IEEE Computer Society Press, McGraw-Hill, 1996 ISBN 0-07-039400-8
Memo	Dr. H. Krebs “Problems of a practicing Surveyor in applying the current draft CENELEC and IEC standards for the testing of safety-critical systems”, Proceedings of Forum European Railway Safety Standards (FERS) 1995

0.4. GLOSSARY

The following definitions arise in part from European and international standards and in part from EEIG ERTMS Users Group terminology. The source relevant to each definition is shown in parenthesis. The acronym EUG is for EEIG ERTMS Usters Group.

Definitions

Application Level	Application Levels of the <i>ERTMS Control/Command</i> system are levels to which a given part of line, or an entire line, or a vehicle can be equipped with <i>ERTMS Control/Command</i> and other equipment. The Application Level and the information available from the signalling system, together with a railway's operating principles, determine the performance level of the ERTMS Control/Command system (EUG)
Assembly	A number of parts or subassemblies or any combination thereof joined together to perform a specific function and capable of disassembly (MIL STD 1388-2B)
Availability	The ability of a product to be in a state to perform a required function under given conditions at a given instant of time or over a given time interval assuming that the required external resources are provided (EN50126)
Availability, Achieved	The ratio between the Up-time and the Total-time of a system or equipment including all repair time (corrective and preventive time), administrative and logistic time (MIL-HDBK-388-1A)
Availability, Intrinsic	Probability that a system or equipment is operating satisfactorily at any point in time when used under stated conditions, where the time considered is operating time and active repair time. Preventive maintenance, administrative and logistic times are excluded (MIL-HDBK-388-1A)
Availability, Operational Conditions, Environmental	see Availability, Achieved The characteristics of the application environment (EUG)
Conditions, Failure	The identification of failures of <i>ERTMS Control/Command</i> functions and the characterisation of

	their effects in term of criticality. <i>ERTMS Control/Command</i> failures are divided in three classes: • immobilising • service • minor (EUG)
Conditions, Maintenance	The maintenance criteria adopted for maintaining the system referred to its Operating Conditions (EUG)
Conditions, Operating	The rated performance required to the system (EUG)
Conditions, System	The conditions under which the system is called to operate, including: • environmental conditions; • operating conditions; • maintenance conditions (EUG)
Contractor	A private sector enterprise or the organisational element of a national railway responsible, against the Project Management, for the system integration in the context of each specific <i>ERTMS</i> application, and for all the Sub-contractors and Suppliers activities within agreed limits specified by the Customer (EUG)
Criticality	A relative measure of the consequence of a failure mode and its frequency of occurrences (MIL-STD-721C)
Customer	The European national railways which intend to apply <i>ERTMS</i> (EUG)
Dependability	The ability of a product to perform one or several required functions under given conditions (EN 50126)
Downtime	The time interval during which a product is in a down state (EN 50126 / IEC50(191))
Context Weakness Parameters	Software function parameters utilised for providing an estimation of the probability that software defects become effective during the function execution and then cause a system failure (EUG)
Environment	The aggregate of all external and internal conditions (such as temperature, humidity, radiation, magnetic and electric fields, shock vibration, etc.) either natural or man made, or self-induced, that influences the form, performance, reliability or survival of an item (MIL-STD-721C)

Equipment/Set	A unit or units and necessary assemblies, or subassemblies and parts connected together or used in association to perform an operational function (MIL-STD-280A)
Error	An error is that part of the system state which is liable to lead to failure. A failure occurs because the system is erroneous (IEC Draft 1508)
ERTMS Control/Command	The European Railway Train Management System, defined from a train onboard point of view, composed by the onboard equipment and all the lineside and trackside equipment necessary for supervising, in real-time, the train operation according to the traffic conditions in reason of the appropriate Level of Application. The <i>ERTMS Control/Command</i> equipment boundary limits, both physical and functional, are defined in accordance to what is stated in the system FRS and SRS (EUG)
Failure	A system failure occurs when the delivered service deviates from the intended service. A failure is the effect of an error on the intended service (IEC Draft 1508)
Failure, Immobilising	An <i>ERTMS Control/Command</i> failure which causes the system to be unable to safely control two or more trains (EUG)
Failure, Minor	A failure which results in excessive unscheduled maintenance and cannot be classified in the above defined failure conditions (EUG)
Failure Rate	The limit, if exists, of the ratio of the conditional probability that the instant of time, T , of a failure of a product falls within a given time interval $(t + \delta t)$ and the length of this interval, δt , when δt tends toward zero, given that the item is in an up state at the start of the time interval (EN50126)
Failure Risk Matrix	Matrix that correlates the failure effects, defined by the function criticality, and the probability of failure appearance, defined by the function weakness class (EUG)
Failure, Service	An <i>ERTMS Control/Command</i> failure which causes the nominal performance of one or more trains to be

	reduced and/or the system to be unable to safely control at most one train (EUG)
Fault	The cause of an error is a fault (e.g. hardware defect, software defect) which resides, temporarily or permanently, in the system (IEC Draft 1508)
Functional Profile	Profile of functions, where profile is a set of disjoint alternatives, each with the probability that it will occur (EUG)
Immobilising Failure	see Failure, Immobilising (EUG)
Interoperability, Operational	The ability of enabling the international safe running of trains on different European networks without: a. having to stop the train at borders; b. changing the engine at borders; c. changing the driver at borders; d. requiring the train driver to perform any other activity different from the standardised ERTMS operation (EUG)
Interoperability, Technical	A subset of operational interoperability, when condition <i>d.</i> is not fulfilled (EUG)
Lifecycle, System	The activities occurring during a period of time that starts when a system is conceived and ends when the system is no longer available for use (IEC draft 1508)
Lifecycle Cost, System	The total cost of acquiring and utilizing a system over its entire life span (MIL-HDBK-388-1A)
Lineside Equipment	see Trackside Equipment (distributed) (EUG)
Logistic Support Resources	The overall resources which are arranged and organised in order to operate and maintain the system at the specified availability level at the required lifecycle cost (EN 50126)
Maintainability	The probability that a given active maintenance action, for an item under given conditions of use can be carried out within a stated time interval when the maintenance is performed under stated conditions and using stated procedures and resources (EN 50126 / IEC50(191))
Maintenance System	A composite of all maintenance resources that must be acquired for maintaining the system throughout its life cycle, including: • spare parts data/documentation/storage;

	• maintenance procedures; • maintenance manuals; • maintenance facilities (power supplies, offices, building of testing centres); • external testing equipment; • special tools; • training of maintenance personnel (EUG)
Maintenance	The combination of all technical and administrative actions, including supervision actions, intended to retain a product in, or restore it to, a state in which it can perform a required function (EN 50126/IEC50(191))
Maintenance, Corrective	The maintenance carried out after fault recognition and intended to put a product into a state in which it can perform a required function (EN 50126 / IEC50(191))
Maintenance Levels	The basic levels of maintenance into which all maintenance activity is divided. (EN 50126)
Maintenance, Preventive	The maintenance carried out at pre-determined intervals or according to prescribed criteria and intended to reduce the probability of failure or the degradation of the functioning of an item (EN 50126/ IEC50(191))
Malfunction	see Failure
Minor Failure	see Failure, Minor (EUG)
Mission Profile	A description of the expected performance of the system in the operational phases of the lifecycle (EN50126)
OnBoard Equipment	see Trainborne Equipment (EUG)
Overlay, Overlaid	A <i>ERTMS Control/Command</i> application where <i>ERTMS Control/Command</i> cooperate with existing systems for accomplishing its mission (EUG)
Part	One piece, or two or more pieces joined together which are not normally subject to disassembly without destruction of designed use (MIL-STD1388-2B)
Project Management	All those activities related to manage the project at the Customer level, including: • system requirements definition; • ensuring the planning, implementation and accomplishment of project related tasks and activities;

	• definition of roles and responsibilities; • definition of resources
	(EUG)
Quality of Service	The collective effect of service performance which determines the degree of satisfaction of a user of a service (EN 50126)
RAM(S) Programme Plan	A document which declares organisation, methodologies, tools and timing for performing the RAM Programme activities (EUG)
RAM(S) Programme	A documented set of time scheduled activities, resources and events serving to implement the organisational structure, responsibilities, procedure, activities, capabilities and resources that together ensure that an item will satisfy given RAM requirements relevant to a given contract or project. (EN 50126 /IEC50(191))
RAM(S) Requirements	The qualitative and quantitative RAM(S) characteristics which the system has to comply with (EUG)
Reliability Growth	A condition characterised by a progressive improvement of a reliability performance measure of an item with time (EN 50126 / IEC50(191))
Reliability	The probability that an item can perform a required function under given conditions for a given time interval (t_1, t_2) (EN 50126 / IEC50(191))
Reliability, Basic	The duration or probability of failure-free performance under stated conditions. Basic reliability terms shall include all item life units (not just mission time) and all failures within the items (not just mission-critical failures at the item level of assembly). Basic reliability requirements is capable of describing item demand for maintenance manpower. The other system reliability parameters employ clearly defined subset of all item life units and all failures (MIL-STD-785B)
Reliability, Mission	The ability of an item to perform its required functions for the duration of a specified mission profile (MIL-STD-721C)
Safety Integrity Level	One of 4 possible discrete levels for specifying the safety integrity requirements of the safety functions to be allocated to the safety related systems. Safety

	Integrity Level 4 has the highest level of safety integrity and Safety Integrity Level 1, the lowest (EN 50126)
Safety Integrity	The probability of a system satisfactorily performing the required safety functions under all stated conditions within a stated period of time (EN 50126)
Schedule Adherence	The ability of a railway system of complying with the schedule of train running (EN 50126)
Service Failure	see Failure, Service (EUG)
Software	Intellectual creation comprising the programs, procedures, rules and any associated documentation pertaining to the operation of a data processing system (ISO9000/3)
Software Component	Software unit 'considered logically indivisible' (for example in the Modula II, Pascal, or C programming languages this term represents one Procedure or one Function) (EUG)
Software Integrity Level	A classification number which determines the techniques and measures that have to be applied in order to minimise residual software faults (prEN50128:1995)
Software Integrity	A measure that signifies the likelihood of software achieving its functions under all stated conditions within a stated period of time (prEN50128:1995)
Software Lifecycle	The activities occurring during a period of time that starts when the software is conceived and ends when software is no longer available for use (prEN50128:1995)
Software Quality	The totality of features and characteristics of a software product that bear on its ability to satisfy stated or implied needs (ISO9000-3)
Spare Parts on Stock	The spare parts which are available on stock (EUG)
Spares	Articles identical to or interchangeable with the end articles on contract which are procured over and above the quantity needed for initial installation for support of a system (MIL-STD-1388-2B)
SQA Plan	The document(s) which formalise(s) all those activities, both technical and managerial, which are necessary to ensure that the software achieves the quality required (prEN50128:1995)

SQA Programme	A documented set of time scheduled activities, both technical and managerial, which are necessary to ensure and to demonstrate, by providing the appropriate evidence, that the software achieves the required level of quality (prEN50128:1995)
Structural Weakness Parameters	Software function parameters utilised for providing an estimation of the probability of injecting defects in the software modules during the software development process (EUG)
Subassembly	Two or more parts which form a portion of an assembly or a unit as a whole, but having a part or parts which are individually replaceable (MIL-STD1388-2B)
Sub-Contractor	A subject responsible, against the Contractor, for providing services or products in the context of <i>ERTMS</i> ; the sub-contractor is also responsible, if applicable, for its Suppliers within agreed limits specified by the Contractor (EUG)
Subsystem	A combination of equipment, units, assemblies, etc., which performs an operational function and is a major subdivision of the system (MIL-STD-721C)
Supplier	Each subject called to directly design and/or produce parts of the <i>ERTMS</i> system (EUG)
System	A composite of equipment, skills, and techniques capable of performing or supporting an operational role, or both. A complete system includes all equipment, related facilities, material, software, services and personnel required for its operation and support to the degree that it can be considered a self-sufficient unit in its intended operational environment (MIL-STD-721C)
System Lifecycle	The activities occurring during a period of time that starts when the system is conceived and ends when system is no longer available for use (EN 50126)
System Quality Plan	The document(s) which formalise(s) all those activities, both technical and managerial, which are necessary to ensure that the system achieves the quality required (EUG)
Trackside Equipment	The equipment with the aim of exchanging information with the vehicle for safely supervising train circulation.

	The information exchanged between track and trains can be either continuous or discontinuous according to the ERTMS Level of Application and to the nature of the information itself. Trackside Equipment is subdivided in two classes: • centralised; • distributed, also called Lineside Equipment (EUG)
Trainborne Equipment	The equipment with the aim of supervising vehicle operation according to the information received from infrastructure installations, from other nonERTMS onboard equipment, from the driver and from the trackside signalling system (EUG)
Unit	An assembly or any combination of parts, subassemblies and assemblies mounted together, normally capable of independent operation in a variety of situations (MIL-STD-280A)
Unsafe state	ERTMS System state due to technical Hazards (excluding the human factor and the external systems), which could lead to an accident.
Validation	Confirmation by examination and provision of objective evidence that the particular requirements for a specific intended use have been fulfilled (IEC draft 1508)
Validation Team	A workgroup jointly chaired by the Customer, responsible for all the specific aspects of the supplying contract, and by EEIG ERTMS Users Group, responsible for the aspects concerning interoperability, and composed by technical and RAMS personnel of EEIG ERTMS Users Group, of the Customer, of the Contractor, of the Sub-contractor(s) and of the Supplier(s) whose aim is to analyse the data resultant from the verification, to require corrective measures where necessary and to provide the RAMS Validation (EUG)
Verification	Confirmation by examination and provision of objective evidence that the specified requirements have been fulfilled (IEC draft 1508)
Weakness	Property of the software function utilised for providing an estimation the function liability to fail, due to the

	software structure and to the environmental conditions (EUG)
Weakness Graph	Qualitative graphical method utilised for giving the ERTMS Control/Command functions the appropriate weakness class, according to the factor, derived by the parameters estimation (EUG)
shall	Means that the relevant verb refers to a requirement
should	Means that the relevant verb refers to a recommendation
can	Means that the relevant verb is permitted
may	Means that the relevant verb is possible

0.5. ACRONYMS AND SYMBOLS

®	Symbol utilised for addressing system requirements along the text of the document
ATO/ATP	Automatic Train Operation/Automatic Train Protection
A_{HW}	Availability of Hardware
$A_{HW,I}$	Mean availability target for Hardware from Immobilising failures per year
$A_{HW,S}$	Mean availability target for hardware service failures per year
$A_{HW,M}$	Mean availability target for hardware minor failures per year
A_{OP}	Operational Availability
A_{TX}	Contribution to availability from Transmission errors
$A_{TX,C}$	Contribution to availability from continuous Transmission errors
$A_{TX,D}$	Contribution to availability from discontinuous Transmission errors
$a_{X,i}$	Allocation factor for the function $f_{X,I}$
BTM	The B alise T ransmission M odule shall interrogate the balises encountered in the track as the trainborne antenna passes them.
CENELEC	Comité Européen de Normalisation Electrotechnique
CMA	Corrective Maintenance Analysis
CTODL	C urrent T ime and O dometer D istribution L ine is designed to provide all modules with frequent up-to-date time, train position, train speed and other train data.
DC	Data Complexity
DR	Defects Reduction
DS	Data Structure Complexity
DT_{HW}	Downtime allowed for hardware failures
$DT_{HW,I}$	Mean downtime allowed for hardware immobilising failures per year
$DT_{HW,S}$	Mean downtime allowed for hardware service failures per year
$DT_{HW,M}$	Mean downtime allowed for hardware minor failures per year

$DT_{TX,C}$	Mean downtime per year due to continuous Transmission errors
$DT_{TX,D}$	Mean downtime per year due to discontinuous Transmission errors
DT_X	Downtime allowed for X criticality failures due to HW
$DT_{X,i}$	Downtime requirement allocated to the function $f_{X,i}$
$DT_{TX,i}$	Downtime due to transmissions which may assume two values: DT_{CON} for continuous TX DT_{DIS} for discontinuous TX or the sum of the two values according to the type of communications required by the function $f_{X,i}$
EB	Emergency Braking
EBC	Emergency Braking Curve
ERTMS	European Rail Traffic Management System
ERTMS-T	It is the trackside of the ERTMS
ETCS	European Train Control System
EUG	EEIG ERTMS Users Group.
EVC	The functions which are located in the E uropean V ital C omputer have very high safety relevance.
FC	Control Flow Complexity
Fd	Defect-caused Failures Frequency
FMEA	Failure Mode and Effect Analysis
FMECA	Failure Mode, Effect and Criticality Analysis
FRACAS	Failure Reporting and Corrective Actions System
FRS	Functional Requirements Specification
FSB	Full Service Braking
FSBC	Full Service Braking Curve
FT	Fault Tolerance Features
FTA	The F ault T ree A nalyses is a graphical method of expressing the logical relationship between a particular failure condition and the failures or other causes leading to the particular failure condition.
$f_{X,i}$	X criticality function i ;
GSM	Global System for Mobile Communications
HF	Human Factor Features
HW	Hardware
$k_{X,i}$	on/off factor: 0 if $f_{X,i}$ does not require communications

1 if $f_{X,i}$ requires communications

I	Immobilising
IL	Integrity Level
ISO	International Standardisation Organisation
LAT	Lineside Active Transducer
LCU	Lineside Control Unit
LNS	Lineside or Trackside Distributed
LPT	Lineside Passive Transducer
LTM	The Loop Transmission Module for track mounted semicontinuouse transmission shall be able to receive telegrams from the track mounted semicontinuous transmission device.
M	Minor
MA	Movement Authority
MC	The aim of the Management Computer is to handle the non-vital trainborne ERTMS functions. To achieve this, the MC must receive the necessary train or track data and process them separately from the safety computations carried out in the EVC.
MMI	All functions which have to be shown to the driver and all choices of the driver are indicated on the Man Machine Interface .
MTBF	Mean Time Between Failures
MTBF-I _{ONB/TRK/LNS}	Mean Time Between Immobilising Failures Onboard/Trackside/Lineside
MTBF-M _{ONB/TRK/LNS}	Mean Time Between Minor Failures Onboard/Trackside/Lineside
MTBF-S _{ONB/TRK/LNS}	Mean Time Between Service Failures Onboard/Trackside/Lineside
MTTR	Mean Time To Repair
MTTRS	Mean Time To ReStore
Nd	Expected Number of Defects
NVF	The vital functions of the radio block center are located in the None Vital computer. These functions have not a very high safety relevance.
OC	Occurrences Level
OCSI	Onboard Complex System Interface
OGA	Onboard GSM Apparatus
OIRT	Onboard Intermittent RX/TX apparatus

ONB	On board
OSI	Onboard System Interface
OSM	Onboard Safe Module
OUI	Onboard User Interface
OUSM	Onboard Unsafe Module
P _{ds}	Probability of having delay due to ERTMS failures
PMA	Preventive Maintenance Analysis
RAM(S)	Reliability Availability Maintainability (Safety)
RBC	Radio Block Centre
Rd	Defects and Failures Reduction
RF	The R ecording F unction is optional. Its purpose is to record all events reported over the ERTMS Bus and to record these together with the time and odometer values at the time of reporting.
RIM	The R adio I nformation M odule shall be transparent to the messages passing it. It shall be able to distinguish between messages for the ERTMS functions and other messages.
RPP	Reliability Programme Plan
S	Service
SIL	Safety integrity level
SQA	Software Quality Assurance
SRS	System Requirements Specification
SSP	Static Speed Profile
SSRS	Sub-system Requirements Specification
STM	The trainborne equipment of the ERTMS must be able to be interfaced with the trainborne equipment of existing train supervision systems. The S pecific T ransmission M odule shall perform a translation function between these systems and the ERTMS.
SW	Software
SZ	Size
SZ1	Estimated Code Length
SZ2	Estimated Code Volume
TCCS	Train Control Command System
TCO	Traction Cut Off
TCOC	Traction Cut Off Curve
TCSI	Trackside Complex System Interface
T _{dn}	Average duration of not delayed trips

T _{dnd}	Average duration of delayed trips
T _{dy}	Acceptable average delay
TF	Time Features
T _{fault}	Time of ERTMS fault condition
TGA	Trackside GSM Apparatus
TIF	The T rain I nterface F unctions are designed to interface a large number of ERTMS functions or individual railway functions that will be technically very dependent on the vehicle type and accordingly cannot be directly connected to the ERTMS trainborne equipment or which are not part of the ERTMS, but which may be useful connected to the ERTMS.
TOF	The T ime and O dometer F unctions have to provide all other modules via a distribution network with frequent messages giving the odometer reading, train velocity, acceleration and a clock reading.
T _{op}	Time of ERTMS correct operation
TRK	Trackside Centralised
TSI	Trackside System Interface
TSM	Trackside Safe Module
T _u	Time of ERTMS unavailability per year
TUSM	Trackside Unsafe Module
V&V	Verification and Validation
VF	The V ital F unctions of the radio block center are located in the Vital computer. These functions have a very high safety relevance.
WC	Weakness Class
X criticality function	I=IMMOBILISING,S=SERVICE,M=MINOR